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GRADE A+**
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning and Preparation using Power Query

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Data Cleaning and Preparation using Power Query

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Problem Statement

A café has received customer transaction data containing several data-quality problems, including missing values, duplicate transactions, blank fields, and invalid entries such as “ERROR” and “UNKNOWN.” These issues make the raw dataset unreliable for accurate reporting and analysis. Manually identifying and correcting these problems in spreadsheets can be difficult, time-consuming, and inconsistent as the volume of data increases.

Therefore, a structured and repeatable data-cleaning process using Power Query is required to identify errors, standardize formats, remove duplicates, handle missing values, and prepare a clean, analysis-ready dataset.

Methodology / Steps Followed

Step 1: Data Collection

- The dataset was obtained from Kaggle and is titled “Cafe Sales – Dirty Data for Cleaning Training.”
- The source file used was dirty_cafe_sales.csv.
- The dataset contains 8 columns:
 - Transaction ID
 - Item
 - Quantity
 - Price Per Unit
 - Total Spent
 - Payment Method
 - Location
 - Transaction Date
- The dataset contains more than 1,000 transaction records.

- The raw data was intentionally created with various data-quality problems such as blank fields, missing values, and invalid entries like “ERROR” and “UNKNOWN.”

Step 2: Data Import

- The CSV file was imported into Power BI Desktop using Get Data → Text/CSV.
- The File Origin was set to 1252: Western European (Windows).
- The delimiter was selected as Comma.
- Data Type Detection was initially kept as Based on first 200 rows.
- The column names and data preview were checked before importing.
- Transform Data was selected to open the dataset in Power Query Editor.

Step 3: Data Profiling

- Column Quality, Column Distribution, and Column Profile were enabled from the View tab in Power Query Editor.
- These features were used to examine the quality and distribution of the data before cleaning.
- Transaction ID was identified as having 100% valid values.
- The Item column contained approximately 90% valid and 10% empty values.
- The Payment Method column contained approximately 68% valid and 32% empty values.
- The Location column contained approximately 60% valid and 40% empty values.
- Quantity, Price Per Unit, and Total Spent were mostly valid, but some errors and blank values were present.
- Invalid placeholder values such as “ERROR” and “UNKNOWN” were identified in multiple columns.

Step 4: Data Cleaning and Transformation

- Invalid entries such as “ERROR” and “UNKNOWN” were replaced with null values using the Replace Values option.
- Missing values in Item, Payment Method, and Location were reviewed and handled according to their relevance to the analysis.

- Some blank entries were categorized as “Not Specified”, while unsuitable records were removed.
- Appropriate data types were assigned to the columns:
 - Transaction ID – Text
 - Item – Text
 - Quantity – Numeric
 - Price Per Unit – Numeric
 - Total Spent – Numeric
 - Payment Method – Text
 - Location – Text
 - Transaction Date – Date
- A new Total Spent value was calculated using:
 - $\text{Total Spent} = \text{Quantity} \times \text{Price Per Unit}$
- This calculation corrected records where the original Total Spent value was missing or incorrect.
- The original Total Spent column was removed and the newly calculated column was renamed Total Spent.
- Duplicate records were removed using Transaction ID.
- Transaction Date values were standardized into a consistent format.
- Text fields were trimmed and cleaned to remove unnecessary spaces and improve consistency.

Step 5: Load and Validation

- After completing the cleaning process, the Column Quality indicators were checked again.
- The cleaned data showed improved validity with reduced errors and empty values.
- The cleaned table was then loaded into the Power BI data model.
- The final dataset was prepared for further analysis, reporting, and visualization.

Output / Results

The screenshot shows the Microsoft Power BI Desktop application. The main window displays a preview of the 'dirty_cafe_sales.csv' file. The data is truncated to the first 200 rows. The interface includes a ribbon with tabs like Home, Insert, Modeling, View, Optimize, and Help, and a sidebar with Visualizations and Filters panes.

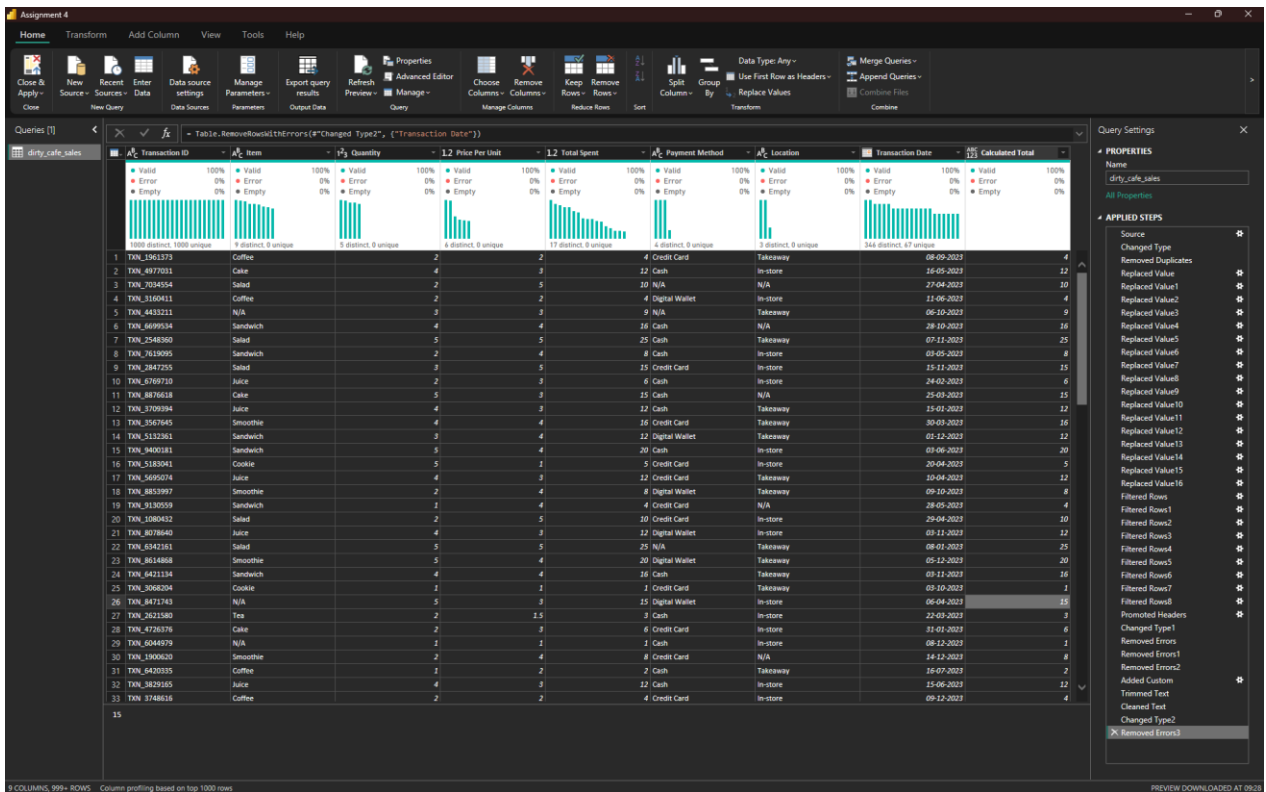
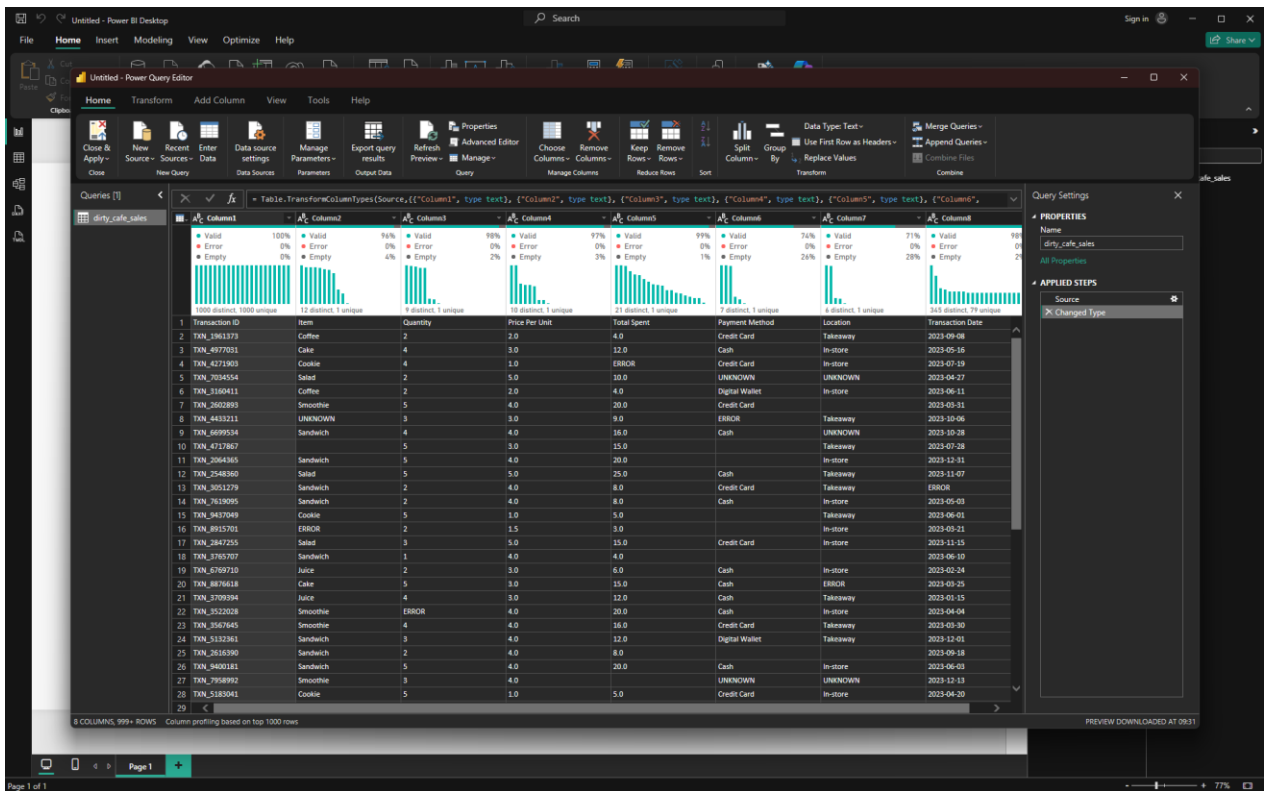
dirty_cafe_sales.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8
Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TAN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
TAN_497021	Cake	4	3.0	12.0	Cash	In-store	2023-05-14
TAN_4271983	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
TAN_3034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
TAN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
TAN_3602893	Smoothie	5	4.0	20.0	Credit Card		2023-03-31
TAN_4433211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway	2023-10-06
TAN_6699534	Sandwich	4	4.0	16.0	Cash	UNKNOWN	2023-10-28
TAN_497021	Salad	5	3.0	15.0		Takeaway	2023-07-24
TAN_2064865	Sandwich	5	4.0	20.0		In-store	2023-12-31
TAN_2548360	Salad	5	5.0	25.0	Cash	Takeaway	2023-11-07
TAN_3051279	Sandwich	2	4.0	8.0	Credit Card	Takeaway	ERROR
TAN_7619095	Sandwich	2	4.0	8.0	Cash	In-store	2023-05-03
TAN_9437049	Cookie	5	1.0	5.0		Takeaway	2023-06-01
TAN_8915701	ERROR	2	1.5	3.0		In-store	2023-03-21
TAN_2847255	Salad	3	5.0	15.0	Credit Card	In-store	2023-11-15
TAN_3763707	Sandwich	1	4.0	4.0			2023-06-10
TAN_4799710	Julie	2	3.0	6.0	Cash	In-store	2023-02-24
TAN_8876618	Cake	5	3.0	15.0	Cash	ERROR	2023-05-25

The data in the preview has been truncated due to size limits.

Buttons: Extract Table Using Examples, Load, Transform Data, Cancel



- The Power Query Editor displayed the original dataset along with its data-quality issues, including blank values and invalid entries such as “ERROR” and “UNKNOWN.”
- The Column Quality, Column Distribution, and Column Profile features helped identify the problematic fields before the cleaning process.
- After applying the transformations, the final Power Query table contained standardized values, corrected numeric fields, cleaned text, standardized dates, and a recalculated Total Spent column.
- The Applied Steps section also recorded the sequence of transformations, making the cleaning process easier to trace and reproduce.

Conclusion

- The experiment successfully demonstrated the use of Power Query for cleaning and preparing café sales data. Various data-quality issues were identified through data profiling and then addressed using Power Query transformation tools.
- Invalid values were replaced, missing data was handled, data types were corrected, duplicate records were removed, dates and text were standardized, and the Total Spent field was recalculated using Quantity and Price Per Unit.
- Overall, Power Query provided a structured, repeatable, and reliable approach to data cleaning compared with manual spreadsheet-based methods. The resulting dataset is suitable for further sales analysis, reporting, and Power BI dashboard development.

Learning Outcomes

- Learned how to use Column Quality, Column Distribution, and Column Profile to identify data-quality issues.
- Gained practical experience in replacing invalid values such as “ERROR” and “UNKNOWN.”
- Learned how to handle missing and null values.
- Understood how to assign suitable data types to different columns.
- Learned how to recalculate derived fields when the original values are missing or incorrect.
- Practiced removing duplicate records using Transaction ID.

- Learned how to standardize date and text formats.
- Understood how Applied Steps can be used to track and reproduce the data-cleaning process.
- Business Insights
 - Payment Method and Location had the highest proportion of missing data, indicating that these fields may require better data collection at the point of sale.
 - The presence of “ERROR” and “UNKNOWN” suggests possible data-entry or system-integration issues that should ideally be corrected at the source.
 - Recalculating Total Spent using $\text{Quantity} \times \text{Price Per Unit}$ helped identify unreliable revenue values.
 - A clean and validated dataset is essential before performing sales, revenue, product, or other business analysis.
 - The cleaned dataset provides a reliable foundation for developing further Power BI reports and dashboards.